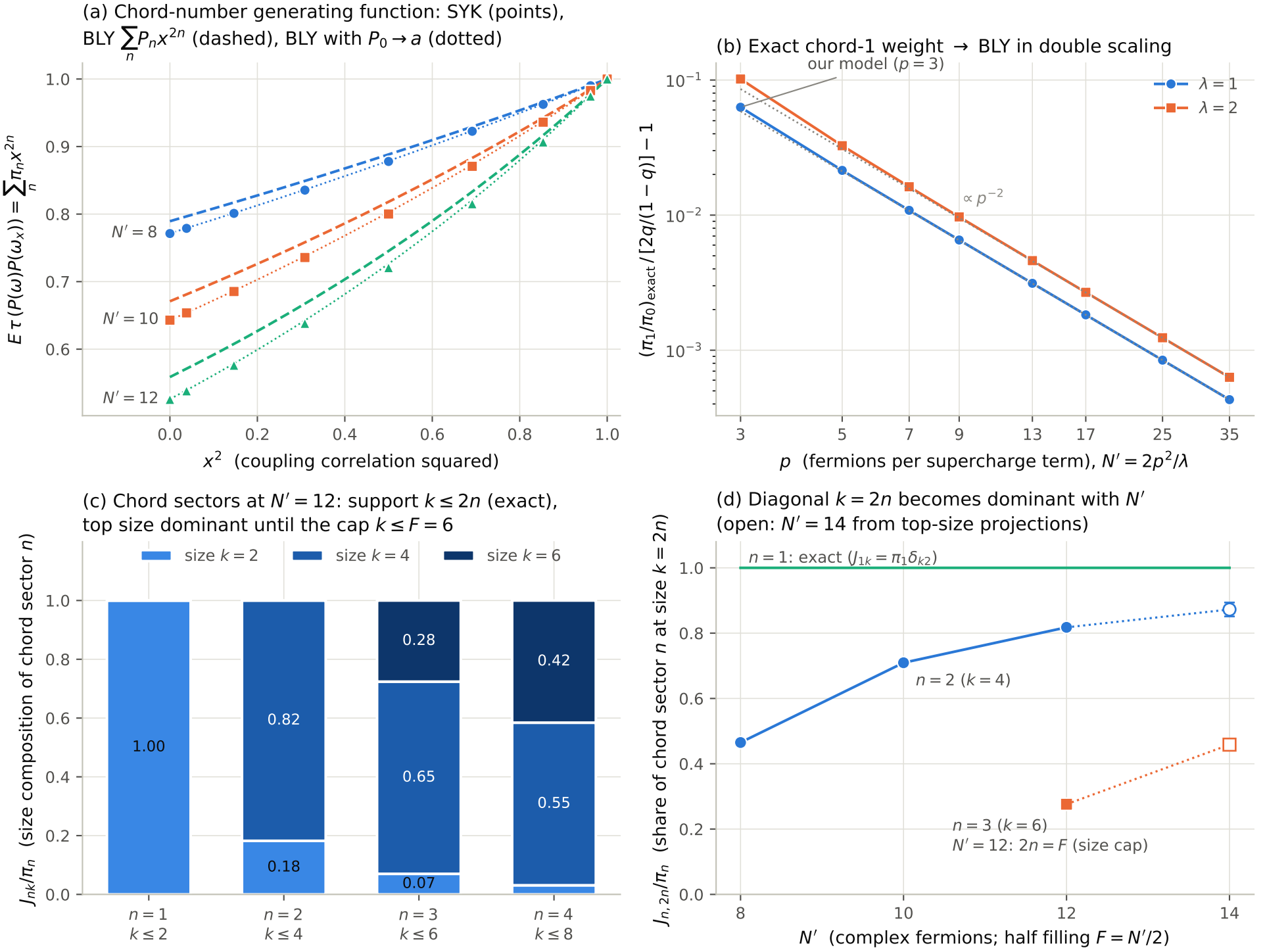
(a) Chord-number generating function: SYK (points), BLY $\sum_n P_n x^{2n}$ (dashed), BLY with $P_0 \to a$ (dotted)
$E\,\tau\,(P(\omega)P(\omega_x)) = \sum_n \pi_n x^{2n}$
$N' = 8$
$N' = 10$
$N' = 12$
x^2 (coupling correlation squared)
(b) Exact chord-1 weight → BLY in double scaling
our model ($p = 3$)
$\lambda = 1$
$\lambda = 2$
$\propto p^{-2}$
$(\pi_1/\pi_0)_{\text{exact}} / [2q/(1-q)] - 1$
p (fermions per supercharge term), $N' = 2p^2/\lambda$
(c) Chord sectors at $N' = 12$: support $k \le 2n$ (exact), top size dominant until the cap $k \le F = 6$
size $k = 2$
size $k = 4$
size $k = 6$
1.00
0.82
0.18
0.28
0.65
0.07
0.42
0.55
J_{nk}/π_n (size composition of chord sector n)
$n = 1$ $k \le 2$
$n = 2$ $k \le 4$
$n = 3$ $k \le 6$
$n = 4$ $k \le 8$
(d) Diagonal $k = 2n$ becomes dominant with N' (open: $N' = 14$ from top-size projections)
$n = 1$: exact ($J_{1k} = \pi_1 \delta_{k2}$)
$n = 2$ ($k = 4$)
$n = 3$ ($k = 6$)
$N' = 12$: $2n = F$ (size cap)
$J_{n,2n}/\pi_n$ (share of chord sector n at size $k = 2n$)
N' (complex fermions; half filling $F = N'/2$)